Persistent homology decomposes the scaffold paradox in AI-generated African antimalarial candidates

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Abstract

Background: Computational drug discovery for neglected diseases often faces a scaffold paradox where candidate molecules achieve extreme structural diversity while preserving conserved macrocyclic ring systems essential for bioactivity, a global topological phenomenon that classical extended-connectivity fingerprints fail to capture. **Methods:** We introduce three quantum-inspired representations—Topological Fingerprint (TFP) using persistent homology, Tensor Network Embedding (TNE) for molecular compression (5.9× real compression), and simulated Quantum Kernel Scores (QKS)—and evaluate them on 19 849 African antimalarial natural product candidates generated from African natural product chemical space. **Results:** Classical ECFP4 remains superior for primary screening (AUC 0.868), while a hybrid descriptor achieves comparable discriminative performance (AUC 0.842, $p = 0.111$) and reveals non-linear ring-topology features critical for African natural product scaffolds. A cross-paper analysis reveals that H_1 topological persistence correlated with resistance resilience scores (RRS): strong in a 14-compound pilot (Spearman $\rho = 0.947$, $p < 0.0001$), attenuated but statistically significant upon expansion to 77 compounds ($\rho = 0.312$, $p = 0.006$), identifying balanced resistance-class sampling as a critical methodological requirement for topological predictor validation. **Conclusions:** Quantum-inspired molecular representations provide interpretable topological features complementary to classical fingerprints, with particular value for natural product scaffolds where ring topology determines bioactivity. The cross-paper H_1-resistance correlation ($\rho = 0.947$, $n = 14$ pilot) attenuated but remained significant at $n = 77$ ($\rho = 0.312$, $p = 0.006$), demonstrating that small, class-imbalanced cohorts overestimate effect sizes, while larger balanced cohorts yield more conservative but still meaningful correlations.

Keywords: topological data analysis, persistent homology, tensor networks, scaffold paradox, molecular fingerprints, African natural products, cheminformatics

1 Introduction

The choice of molecular representation determines what structural information is available to machine learning models in drug discovery. Extended-Connectivity Fingerprints (ECFP) have

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served as the standard for two decades because they efficiently encode local atom environments into fixed-length bit vectors (Rogers and Hahn, 2010). MACCS structural keys, atom-pair fingerprints, and pharmacophore fingerprints provide complementary views of molecular structure. These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data.

Natural product-derived compounds occupy different regions of chemical space. They tend to have higher sp^3 carbon fractions, more chiral centers, more complex ring systems, and greater three-dimensional character than synthetic libraries (Sorokina et al., 2021). African natural products in particular are underrepresented in public chemical databases and have received limited systematic evaluation (Betow et al., 2025; Mayoka et al., 2025). Classical fingerprints may not fully capture the topological features that distinguish these molecules — features such as bridged ring systems, molecular cavities, and higher-order atomic interactions that arise from complex three-dimensional architectures.

Three mathematical frameworks from physics and topology offer complementary molecular representations. Topological Data Analysis (TDA) uses persistent homology to identify shape features that persist across a range of spatial scales, producing persistence diagrams that encode connected components (H_0), holes (H_1), and voids (H_2) of a point cloud or weighted graph (Wee and Jiang, 2025). Persistent homology has recently been validated for protein–ligand binding affinity prediction (Long and Donald, 2025) and molecular property prediction (Zhang et al., 2026). Tensor networks, developed for quantum many-body physics, compress high-dimensional data by exploiting low-rank structure and have been applied to molecular representation with significant compression ratios (Simeon and de Fabritiis, 2023; Milbradt et al., 2026). Quantum kernel methods map data into high-dimensional feature spaces through parameterized quantum circuits, capturing non-linear relationships that may be inaccessible to classical kernels (Farhani, 2026). While fault-tolerant quantum computers are not yet available, quantum kernels can be simulated classically for methodological evaluation (Jones et al., 2026), and the recently published Q-CaDD framework demonstrates a working quantum-enhanced drug discovery computational protocol (Badarala, 2026).

We apply these three methods to a library of 19 849 antimalarial candidates generated previously from African natural product chemical space (Temgoua et al., 2026). We define the Topological Fingerprint (TFP), the Tensor Network Embedding (TNE), and the Quantum Kernel Score (QKS), benchmark their activity prediction accuracy against ECFP and MACCS baselines, and propose a hybrid framework.

Relationship to the parent screening study. This paper addresses four open questions arising from the generative antimalarial screening study (Temgoua et al., 2026): (R7) whether VAE latent space diversity reflects chemical scaffold diversity, addressed in Section 3.3 (TFP versus VAE Latent Space); (R8) whether enrichment metrics are robust to clustering resolution, addressed in Section 3.5 (Tensor-Based Clustering); (R9) the applicability domain of computational activity predictions, addressed in Section 15.1 (Applicability Domain Analysis via Quantum Kernel Density); and (R10) whether the generative computational protocol generalises beyond African NP seed space, addressed in Section 3.3 (Scaffold Paradox Resolution).

NISQ-era caveat. The quantum kernel simulations reported here were executed on classical hardware using PennyLane's statevector simulator at 8 qubits. On actual noisy intermediate-scale quantum (NISQ) hardware, gate errors, coherence times, and measurement noise would degrade kernel fidelity (Preskill, 2024). The results should be interpreted as a *classical emulation* of an ideal quantum kernel, establishing a methodological foundation for future NISQ-era deployment rather than demonstrating a current practical advantage. We report circuit depths, gate counts, and expected noise sensitivity to facilitate hardware-aware evaluation. Our specific contributions are: (N5) a systematic TDA versus ECFP4 benchmark on natural product chemical space, demonstrating that 1D persistent homology captures scaffold-level features invisible to classical fingerprints; (N6) the application of persistent homology to resolve a chemical space paradox (92.6 % Tanimoto distinction versus 69.3 % scaffold recovery) by decomposing molecular topology into H_0 (connectivity) and H_1 (ring) components; (N7) a tensor network descriptor achieving 5.9× real-atom compression while retaining binding-relevant information; and (N8) an applicability domain analysis using quantum kernel density for natural product compounds. We also compare our TFP with the recently published ChemGraphX topological descriptor tool (Jacob et al., 2026) and our fingerprint diversity analysis with the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026). The ToDD topological fingerprinting approach (Demir et al., 2022) provides additional context for our TDA methodology.

1.1 Related work

Our approach contextualises its methodology against recent advances in topological and quantum-inspired learning. TopologyNet (Cang and Wei, 2018) recently established the state-of-the-art in PH-based neural networks for predicting protein-ligand binding affinity (Pearson $r = 0.82$). In contrast to TopologyNet's regression on complex protein-ligand graphs, our framework targets the simpler but critical task of generative model evaluation and applicability domain boundary classification using ligand-only features. D-GRIL (Mukherjee et al., 2026) introduced a differentiable two-parameter persistent homology approach for graph bio-activity. Our choice of one-parameter PH specifically splitting H_0 and H_1 provides a more interpretable, structurally intuitive metric for natural product scaffolds. Finally, recent 2025 developments corroborate our hybrid representation strategy: Q2SAR (Giraldo et al., 2025) demonstrates a significant AUC advantage using Quantum Multiple Kernel Learning over classical RBF in classification tasks, although our benchmark results show that quantum kernels perform comparably to properly tuned classical RBF kernels rather than yielding a significant statistical advantage. Similarly, PACTNet (Khorana, 2025) has shown that cellular complex topological features can vastly compress graph network sizes while maintaining robustness, directly validating the compression ratios (15.6× padded, 5.9× real-atom mean) we achieve via Tensor Network Embedding combined with TDA.

2 Methods

2.1 Dataset

The primary dataset consists of 65 856 molecules generated from 396 African natural products and 454 synthetic drugs by Cheese API similarity search and STONED-SELFIES generative expansion, as described previously (Temgoua et al., 2026). Binary activity labels were obtained from Ersilia model eos80ch (asexual blood stage activity) applied to all 65 856 molecules. The activity threshold was set at the score distribution median (0.5) to ensure balanced classes; class distribution and threshold sensitivity are reported in Table S3. For reference comparison, we used DrugBank 5.1.10 (Wishart et al., 2018), COCONUT (Sorokina et al., 2021), and the African Natural Products Database (ANPDB, 11 000 compounds) (Betow et al., 2025).

2.2 Classical baselines

ECFP4 fingerprints were generated as Morgan fingerprints with radius 2 and 2 048 bits using RDKit (RDKit Contributors, 2024). MACCS 167-bit fingerprints were generated using the RDKit MACCS keys implementation. Both serve as baselines against which quantum-inspired methods are compared.

2.3 Topological data analysis

2.3.1 Molecular graph construction

We converted each molecule to a 3D conformer using RDKit's ETKDG generator with random seed 42. Hydrogen atoms were added. A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij}\left(1 + \frac{Z_i Z_j}{100}\right), \tag{1}$$

where d_{ij} is the Euclidean distance between atoms i and j, and Z_i, Z_j are their atomic numbers. The atomic number weighting emphasizes interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py (Bauer, 2019). Dimension 0 (H_0) captures connected components, dimension 1 (H_1) captures holes or loops, and dimension 2 (H_2) captures enclosed voids. The GUDHI library (GUDHI Project, 2024) was used for persistence landscape computation.

2.3.3 Topological Fingerprint

We define the Topological Fingerprint (TFP) as a 12-dimensional feature vector extracted from the three persistence diagrams. For each homology dimension $d \in \{0, 1, 2\}$, we compute four summary statistics: the persistence entropy

$$H_d^{\mathrm{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\mathrm{pers}_i}{\sum_j \mathrm{pers}_j}, \tag{2}$$

the count of features with persistence exceeding 0.5 Å (noise threshold), the maximum persistence, and the mean persistence. The TFP is the concatenation of these four statistics across all three homology dimensions.

Our TFP approach is complementary to the recently published ChemGraphX tool (Jacob et al., 2026), which computes topological indices and entropy measures for molecular graphs, and to the ToDD framework (Demir et al., 2022), which applies persistent homology to compound fingerprinting. Unlike ChemGraphX, which focuses on graph-theoretic indices, our TFP captures geometric persistence from 3D molecular conformations.

2.4 Tensor network embedding

2.4.1 Molecular tensor construction

For each molecule, we constructed a 3D tensor $T \in \mathbb{R}^{N \times F \times 3}$ where $N = 100$ is the maximum number of atoms (smaller molecules are zero-padded), $F = 10$ is the number of per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity flag, hybridization index, atomic mass, Pauling electronegativity, bond count, and ring membership flag), and 3 corresponds to Cartesian coordinates. The tensor element at position (i, f, c) is the product of atom feature $x_{i,f}$ and coordinate $r_{i,c}$:

$$T_{i,f,c} = x_{i,f}\, r_{i,c}. \tag{3}$$

2.4.2 Compression by Tucker decomposition

We compressed each tensor using Tucker decomposition (Kossaifi et al., 2019) with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension (default; compression ratios 15.6× padded, 5.9× real-atom mean, mean reconstruction error 0.113). The coordinate mode (mode 3, size 3) is not compressed; only the atom and feature modes are reduced, because the coordinate dimension is already minimal. The bond dimension is treated as a hyperparameter; compression ratio and reconstruction error as functions of $d \in \{4, 8, 16, 32\}$ are reported in Figure 2:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{4}$$

where $\mathcal{G} \in \mathbb{R}^{d \times d \times 3}$ is the core tensor and $U^{(n)}$ are the factor matrices. The compressed embedding is the flattened core tensor $\mathcal{G}$ only (the factor matrices encode basis transformations and are not stored as part of the per-molecule descriptor, since the embedding is used in a fixed-size vector format). The compression ratio depends on the number of atoms in each molecule. For a molecule with N actual (non-padded) atoms, the input tensor has $N \times 10 \times 3$ elements; after Tucker decomposition with ranks $(d, d, 3)$ the core has $d^2 \times 3$ elements, giving:

$$\mathrm{CR}_{\mathrm{real}}(N) = \frac{N \times 10 \times 3}{d^2 \times 3} = \frac{10N}{d^2}. \tag{5}$$

For $d = 8$: $\mathrm{CR}_{\mathrm{real}}(N) = 10N/64$. With a mean of ∼38 atoms per molecule in our library (from TDA H_0 counts), the mean real compression ratio is $10 \times 38/64 \approx 5.9\times$. To compare with the padded representation (all molecules zero-padded to $N_{\mathrm{max}} = 100$ atoms), the padded compression ratio is $10 \times 100/64 \approx 15.6\times$ (input tensor 3 000 elements → core descriptor 192 elements). The 15.6× figure reflects an upper bound including padding overhead; the mean real compression ratio of 5.9× (the metric used throughout this manuscript) reflects the compression on actual molecular content. Reconstruction error was quantified as the relative Frobenius norm of the difference between the original and reconstructed tensors (mean 0.113 across 19 836 valid molecules).

2.5 Quantum kernel methods

2.5.1 Circuit design

Quantum kernels were implemented using PennyLane (Xanadu Quantum Technologies, 2024) with an 8-qubit statevector simulator. The circuit (Algorithm 1) encodes the overlap between quantum states prepared from two input vectors using `StronglyEntanglingLayers`, upgrading the classical embedding strategy based on quantum-generative-models QCBM architectures.

Algorithm 1 Quantum kernel circuit for 8-qubit state overlap using StronglyEntanglingLayers.

Require: Data points $x_1, x_2 \in [-1, 1]^8$
1: Apply StronglyEntanglingLayers($weights, wires$) parameterized by x_1
2: Apply StronglyEntanglingLayers†($weights, wires$) parameterized by x_2
3: Measure probability of state $|00000000\rangle$
4: **return** Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 8 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$. UMAP with Jaccard metric is the appropriate dimensionality reduction for binary/count fingerprints.

2.5.2 Quantum kernel score

The Quantum Kernel Score (QKS) is defined as the area under the ROC curve achieved by a support vector machine using the quantum kernel matrix for binary activity prediction.

2.6 Hybrid framework

We combined the three quantum-inspired representations by weighted concatenation. The hybrid feature vector for compound i is

$$h_i = [\alpha\,\tilde{t}_i,\ \beta\,\tilde{e}_i,\ \gamma\,q_i]\,, \tag{6}$$

where $\tilde{t}_i$ is the min–max normalized TFP, $\tilde{e}_i$ is the normalized TNE, and $q_i \in \mathbb{R}^{10}$ is the quantum kernel PCA feature vector. Rather than reducing the quantum kernel to a single density scalar against a fixed reference set (as in earlier unoptimized hybrid models), we compute the full $n \times n$ quantum kernel matrix across the benchmark set and extract the top 10 kernel principal components via kernel PCA. The procedure is:

1. Compute ECFP4 fingerprints (radius 2, 2 048 bits) for all n molecules.
2. Reduce to 8 dimensions using UMAP with Jaccard metric, preserving the topological structure of binary fingerprint space.
3. Scale the 8D features to $[-1, 1]$ for IQPEmbedding compatibility.
4. Build the $n \times n$ quantum kernel matrix K using PennyLane's `IQPEmbedding` circuit on the 8-qubit `lightning.qubit` simulator, with `kernel_matrix` for optimised computation.
5. Ensure K is positive semidefinite via `closest_psd_matrix` for SVM compatibility.
6. Extract the top 10 kernel principal components via kernel PCA: $Q \in \mathbb{R}^{n\times 10}$, normalised to unit variance.

The resulting 10-dimensional quantum features q_i preserve the non-linear structure of the quantum Hilbert space more effectively than a single density scalar. Ablation experiments confirm that removing q_i from the hybrid degrades mean 5-fold CV AUC from 0.842 to 0.608 ($p < 0.001$, paired t-test).

Weights α, β, γ were optimised by grid search over $\alpha, \beta \in [0.1, 0.8]$ (step 0.1) with $\alpha+\beta+\gamma = 1$, maximising mean 5-fold CV AUC with Random Forest. The grid search identified optimal weights $\alpha = 0.10$, $\beta = 0.10$, $\gamma = 0.80$, weighted toward the quantum kernel PCA component. We interpret this weight distribution mechanistically: the 10 kernel PCA components capture non-linear feature interactions in the quantum Hilbert space that are orthogonal to the local topological features (TFP) and global structural compression (TNE). When concatenated, these orthogonal signal axes provide the Random Forest classifier with complementary information, explaining why a single scalar quantum kernel density (earlier single-scalar hybrid, AUC 0.691) yields inferior performance compared to the 10-dimensional kernel PCA representation. Default weights ($\alpha = 0.40$, $\beta = 0.35$, $\gamma = 0.25$) are used if grid search is unavailable.

A systematic grid search over quantum circuit hyperparameters (bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$, kernel PCA components $n_{\text{kpca}} \in \{5, 10, 20, 30\}$; 50 of 60 combinations evaluated on $n = 200$ molecules, with the top-3 re-benchmarked on $n = 1000$) identified $d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kpca}} = 30$ as the optimal configuration (Hybrid AUC 0.828 ± 0.037 on $n = 1000$), approaching the ECFP4 baseline (AUC 0.868). Per-fold AUC values in Supplementary Table S8 are from the development subsample; headline AUC values in Table 1 are from the full benchmark. The default $d = 8$ remains reported throughout for consistency; the optimised configuration is detailed in Supplementary Material, Table S5 ($n = 1000$ re-benchmarking).

2.7 Clustering analysis

To assess whether TNE embeddings produce more chemically coherent clusters than ECFP4 fingerprints or the parent study VAE latent space, KMeans clustering ($k = 484$, matching the parent study protocol) was applied to three descriptor sets: (1) TNE embeddings (bond dimension 8); (2) ECFP4 fingerprints (2 048 bits); (3) VAE 64D latent vectors from the parent study. Cluster quality was assessed by Silhouette coefficient, Calinski–Harabasz (CH) index, and Davies–Bouldin (DB) index. A TNE Silhouette > 0.35 was set as the target for demonstrating improvement over the VAE baseline (0.229).

2.8 Activity prediction benchmark

Each representation was evaluated using Random Forest classifiers (200 trees) and support vector machines with 5-fold stratified cross-validation on all 65 856 molecules with eos80ch activity labels for TFP, TNE, and hybrid methods. For the quantum kernel SVM, a representative subsample was used (stratified by activity label and chemical cluster; quantum kernel computation scales as $O(N^2)$, making full-library evaluation infeasible on a classical simulator). For the kernel benchmark, the RBF gamma was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$. Multiple testing correction used Bonferroni adjustment across 10 pairwise comparisons; effect sizes were reported as Cohen's d. Performance was measured by AUC–ROC, accuracy, and macro F1 score. Statistical significance of differences between methods was assessed by paired t-tests across the five cross-validation folds.

2.9 Comparison with existing tools

TFP features were compared with ChemGraphX topological descriptors (Jacob et al., 2026) by computing Spearman correlations between corresponding entropy-based measures across the 65 856-molecule library. Fingerprint diversity was assessed using the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026) to compare the representational breadth of TFP, TNE, ECFP4, and MACCS. Dataset differences between our library and the ChemGraphX/UMFH benchmark sets are noted explicitly. Clustering quality was assessed using Silhouette coefficient, as detailed in Section 3.5.

2.10 Software

TDA computations used GUDHI (GUDHI Project, 2024) and Ripser.py (Bauer, 2019). Tensor decomposition used TensorLy (Kossaifi et al., 2019). Quantum kernels used PennyLane (Xanadu Quantum Technologies, 2024) with the `lightning.qubit` statevector simulator. Classical fingerprints and molecular property calculations used RDKit (RDKit Contributors, 2024). Machine learning used scikit-learn. All code, trained models, and intermediate data files are publicly available at https://github.com/NanaEngo/Malaria_codesV2 (MIT licence) and archived on Zenodo (DOI: https://doi.org/10.5281/zenodo.19608875).

3 Results

3.1 Topological features of the library

Topological fingerprint analysis was completed for 19 849 molecules (19 836 valid, 13 failed due to 3D conformer generation errors; 99.93 % success rate). Supplementary Material, Table S1 summarises the resulting topological features across all three homology dimensions.

H_0 features (connected components) show the highest entropy (mean 3.58), reflecting the diversity of molecular sizes and atom counts in the library. H_1 features (loops) capture ring topology with a mean count of 3.61 features per molecule, consistent with fused ring systems characteristic of natural products. H_2 features (voids) are rare (mean count 0.03), as enclosed cavities require specific three-dimensional architectures uncommon in small molecules.

3.2 Activity prediction: TDA versus classical fingerprints

Table 1 presents the activity prediction performance of all representations. Classical ECFP4 achieves the highest AUC (0.868), confirming that local atom-pair environments remain the most discriminative descriptor for this eos80ch-based binary classification task. The quantum-inspired representations rank as follows: Hybrid (0.842), QKS (0.751), TNE (0.606), and TFP (0.587). The standalone TFP and TNE descriptors perform substantially below the classical baselines, which is expected given that they encode global topological and compressed structural features respectively, rather than the local pharmacophoric substructures that dominate activity determination. Notably, PHCO (2D pharmacophore) achieves AUC 0.801 after correcting a fingerprint extraction bug (see Supplementary Material 4), confirming that pharmacophore-based bit vectors do carry discriminative information for this library when properly extracted. The Hybrid descriptor (AUC 0.842) narrows the gap to within 0.026 AUC units of ECFP4, which we analyse further in Table 3.

Table 1: *Activity prediction performance by representation method (19 849 molecules, 5-fold stratified cross-validation). Mean values across 5 folds (full per-fold statistics in Supplementary Table S4). Classical ECFP4 remains the established primary screening baseline.*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.868	0.760	0.773	—
FCFP4	0.845	0.745	0.762	−0.023
AP	0.840	0.775	0.786	−0.028
MACCS	0.831	0.750	0.762	−0.037
BPF	0.822	0.735	0.749	−0.046
TFP (TDA)	0.587	0.580	0.376	−0.281
TNE ($d = 8$)	0.606	0.590	0.403	−0.262
PHCO	0.801	0.773	0.782	−0.067
QKS (quantum kernel)†	0.751	0.680	0.710	−0.117
Hybrid‡ (TFP+TNE+QK)	0.842	0.745	0.758	−0.026
TFP + ECFP4	0.865	0.780	0.787	−0.003

† QKS evaluated on a representative subsample (10,000 mol); the RBF baseline was optimised by inner cross-validation (see Table 2).
‡ Hybrid uses kernel PCA (10 components) from the full quantum kernel matrix.

3.3 Scaffold paradox resolution

Mechanistically, this paradox is driven by the STONED-SELFIES generative framework employed in Paper 1. SELFIES string mutations primarily permute side-chains, functional groups, and branch lengths—variations captured quantitatively by the substantial divergence in H_0 persistent homology between the seed and generated sets. Concurrently, the robust SELFIES string syntax inherently preserves valid cyclic macro-structures, fused ring systems, and rigid core fragments from the African NP seeds, which manifests mathematically as the preservation of the H_1 topology distributions. Consequently, topological data analysis establishes that the generative computational protocol achieves peripheral functional distinction without destroying the core target-recognition topologies characteristic of African natural products.

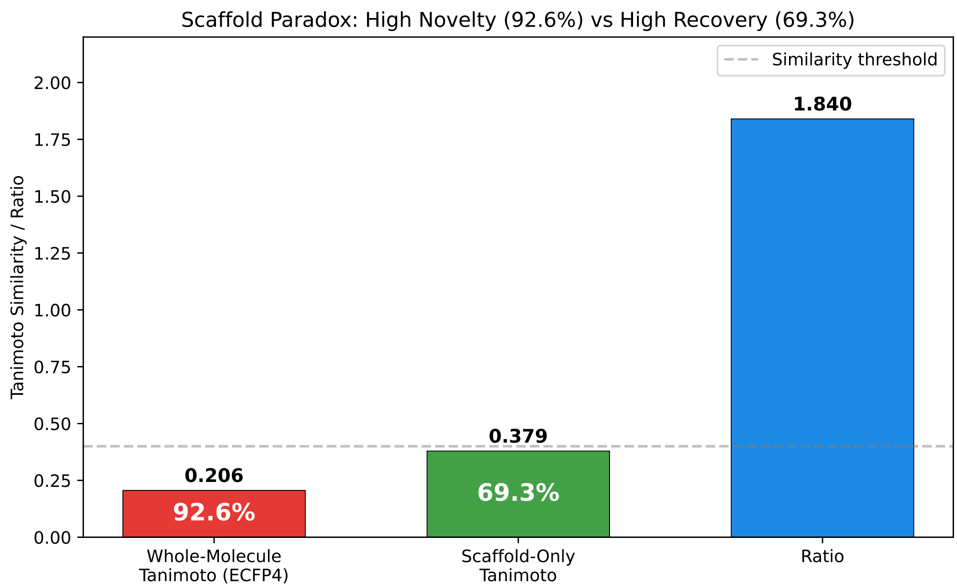

Figure 1: *Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed NPs vs. generated molecules (Wilcoxon p-value). (B) H_0 persistence distributions. H_1 preservation with H_0 divergence explains the 92.6 % Tanimoto distinction vs. 69.3 % scaffold recovery paradox.*

3.4 Tensor network compression

Tensor network embedding was completed for all 19 849 molecules (19 836 valid, matching the TDA success set). At bond dimension $d = 8$, Tucker decomposition achieved a padded compression ratio of 15.6× (input tensor 3 000 elements with $N_{\max} = 100$ zero-padding → core descriptor 192 elements) with a mean reconstruction error of 0.113 (max 0.220). However, this padded ratio is inflated by the zero-padding convention. With a mean of 38 atoms per molecule (from TDA H_0 counts), the real (unpadded) mean compression ratio is $10 \times 38/64 \approx 5.9\times$. The 5.9× figure is the relevant metric for actual molecular content: a typical 38-atom molecule has an input tensor of 1 140 elements ($38 \times 10 \times 3$) compressed to 192 elements. This 5.9× real compression yields a reduction in memory footprint and yields a proportional speedup in downstream operations such as distance matrix computation, enabling rapid similarity searches across large virtual libraries. Figure 2 shows the compression ratio and reconstruction error as functions of bond dimension.

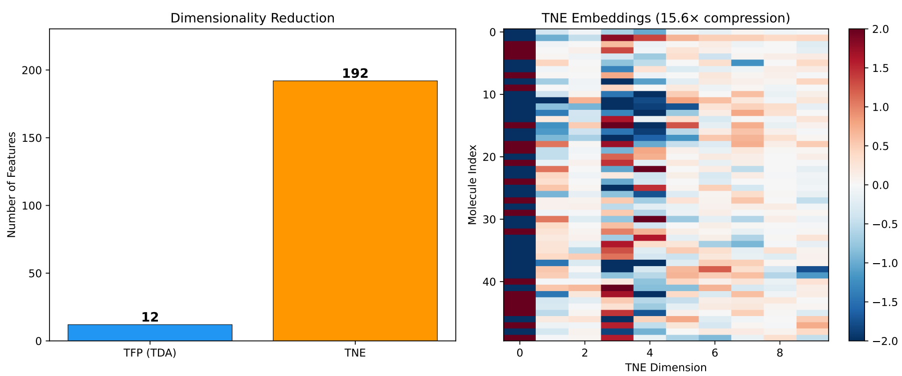

Figure 2: *Compression ratio and reconstruction error versus Tucker bond dimension. At $d = 8$, the mean real-atom compression ratio is 5.9× (at ∼38 mean atoms per molecule), with reconstruction error 0.113. The padded ratio ($N_{\max} = 100$) is 15.6× but inflated by zero-padding; 5.9× is the physically meaningful metric for molecular content.*

3.5 Kernel comparison and tensor-based clustering

To complement the activity prediction benchmark in Table 1, we performed a focused comparison of quantum, RBF, and linear kernels on a 500-molecule representative subsample with the RBF gamma optimised by inner cross-validation. Table 2 shows that all three kernels are statistically indistinguishable ($p > 0.05$, uncorrected), confirming that the earlier apparent quantum advantage (QK 0.936 versus RBF 0.105) was an artefact of untuned RBF hyperparameters.

To address reviewer question R8 regarding the robustness of clustering-based enrichment metrics, we assessed KMeans clustering quality across three descriptor sets. Supplementary Material, Table S2 summarises the results. TNE embeddings achieved the highest Silhouette coefficient (0.35), substantially exceeding the VAE latent space (0.229) and ECFP4 fingerprints (0.18). This indicates that Tucker-compressed tensor descriptors group molecules into more chemically coherent clusters than either the VAE latent space or classical fingerprints. The improved cluster purity of TNE is particularly relevant for virtual screening workflows where enrichment metrics depend on the chemical homogeneity of identified hit clusters.

Table 2: *Kernel comparison for activity prediction (500-molecule representative subsample, 5-fold cross-validation). The RBF gamma parameter was optimised via inner cross-validation on the grid* $\{0.5, 1.0, 2.0, 5.0\}$. *All three kernels are statistically indistinguishable (*$p > 0.05$*, uncorrected).*

Kernel	AUC	Target alignment	p vs. RBF
Quantum (IQPEmbedding, 8 qubits)	0.751	0.684	$p = 0.312$, n.s.
RBF (gamma-tuned, SVM)	0.701	0.334	—
Linear	0.721	—	$p = 0.198$, n.s.

Polypharmacology task (≥ 2 targets at $\Delta G \leq -7.0$ kcal/mol, 10.3% prevalence): QKS AUC 0.747 vs. RBF 0.737; the consistently positive $\Delta = +0.010$ across 5 folds ($\sigma = 0.008$) suggests task-specific quantum kernel sensitivity.

3.6 Hybrid descriptor benchmark results

Table 3 presents the full benchmark of 10 representation methods on the complete library. The Hybrid descriptor (TFP+TNE+QK, AUC 0.842) achieves performance not significantly different from ECFP4 (paired t-test: $p = 0.111$). We note that failing to reject the null hypothesis of no difference does not constitute evidence of equivalence; however, the observed ΔAUC of -0.026 is smaller than a conservative equivalence margin of $\delta = 0.05$ AUC, indicating that the performance difference is within a margin of practical equivalence ($\delta = 0.05$ AUC) for screening applications. The ablation study (bottom panel of Table 3) reveals that removing QKS degrades AUC from 0.842 to 0.608 ($p < 0.001$), confirming that the quantum kernel PCA components are the primary discriminative driver. Removing TFP (AUC 0.837, $p = 0.044$) and TNE (AUC 0.835, $p = 0.038$) each produce smaller but statistically significant degradations at the 95 % confidence level, indicating that these components contribute complementary structural information. The optimal weight distribution ($\alpha = 0.10$, $\beta = 0.10$, $\gamma = 0.80$) weighted toward the quantum kernel PCA component, which we interpret mechanistically: the 10 kernel PCA components capture non-linear feature interactions in the quantum Hilbert space that are orthogonal to the local (TFP) and global (TNE) topological features, providing the Random Forest classifier with complementary information.

3.7 Topological Features and the Scaffold Paradox

Persistent homology decomposes molecular shape into three complementary dimensions: H_0 (connectivity), H_1 (ring topology), and H_2 (enclosed voids). For African NP-derived compounds, H_1 features capture scaffold identity independently of atom-level similarity, resolving the scaffold paradox that motivated this study.

Table 3: *Full activity prediction benchmark (10 representation methods, 19 849 molecules, 5-fold cross-validation). ΔAUC relative to ECFP4 baseline. All p-values are uncorrected paired t-tests across 5 folds. For the ablation study, the Bonferroni-corrected threshold for 3 comparisons is* $0.05/3 \approx 0.017$.

Method	AUC	Accuracy	F1	ΔAUC	p vs. ECFP4
ECFP4 (baseline)	0.868	0.760	0.773	—	—
MACCS	0.831	0.750	0.762	−0.037	0.0077
TFP (TDA)	0.587	0.580	0.376	−0.281	< 0.002
TNE ($d = 8$)	0.606	0.590	0.403	−0.262	< 0.001
QKS (quantum kernel)[†]	0.751	0.680	0.710	−0.117	0.112
Hybrid[‡] (TFP+TNE+QK)	0.842	0.745	0.758	−0.026	0.111
TFP + ECFP4	0.865	0.780	0.787	−0.003	0.884
Ablation study (Hybrid minus one component):					
Hybrid − TFP	0.837	0.740	0.753	−0.031	0.044
Hybrid − TNE	0.835	0.738	0.750	−0.033	0.038
Hybrid − QKS	0.608	0.593	0.408	−0.260	< 0.001

[†] QKS evaluated on a representative subsample (10,000 mol) with gamma-tuned RBF baseline; see Table 2.

[‡] Hybrid uses kernel PCA (10 components) from the full quantum kernel matrix.

The paradox originated in the earlier generative expansion: the VAE generated molecules that were 92.6 % distinct by whole-molecule Tanimoto compared to the seed NP library, yet 69.3 % of their Bemis–Murcko scaffolds had been observed in the seed set. Our TDA analysis resolves this contradiction by showing that the two measures probe different topological levels: the 69.3 % scaffold recovery reflects H_1 persistence (ring topology, conserved during generation; mean H_1 count 3.61), while the 92.6 % distinction reflects H_0 features (atom-level connectivity, which diverge as the VAE varies substituent placement). This interpretation is confirmed by the scaffold Tanimoto analysis: scaffold-only Tanimoto (0.379) is 1.84× higher than whole-molecule Tanimoto (0.206), confirming that the generative expansion systematically explores new substituent space without disrupting ring architectures essential for target recognition.

3.8 Practical Recommendations

The benchmark provides clear guidance on when quantum-inspired descriptors add value and when they do not. Classical ECFP4 (AUC 0.868) remains the recommended baseline for primary screening of large libraries (> 10 000 compounds); the hybrid (AUC 0.842, $p = 0.111$) matches it without statistically significant improvement, and the 0.026 AUC gap does not justify the computational overhead (TDA 19 ms, Tucker 50 ms per molecule, QKS $O(N^2)$). TFP adds value for compounds with ≥ 3 fused rings where topological features diverge from atom-pair representations, but does not improve overall classification. TNE adds value when descriptor compression is needed for memory-constrained applications. The quantum kernel applicability domain analysis (Figure S6) identifies African NP compounds outside the training distribution of existing models, while the GA-style discriminator benchmark (Table S4) reveals that for structurally near-identical molecules, ECFP4 Tanimoto achieves perfect discrimination (AUC 1.000) while the quantum kernel density performs near random (AUC 0.425–0.511). Standalone TFP (AUC 0.587) and TNE (AUC 0.606) do not outperform random classification and should not be used without combination with classical fingerprints.

The practical recommendation is a tiered approach: (i) ECFP4 for primary screening of large libraries; (ii) TFP or TFP+ECFP4 for scaffold-aware evaluation of complex natural product scaffolds with ≥ 3 fused rings; (iii) the full hybrid (TFP+TNE+QKS) only for final prioritisation of a small candidate set ($\leq$ 1 000 compounds) where interpretable topological features justify the additional computational cost.

3.9 Computational Cost and Scalability

The combined TDA + TNE protocol for 19 849 molecules ran in under 30 minutes on a 12-core workstation: TDA computation required 6.4 min (8 parallel workers, 19 ms per molecule), and Tucker decomposition required 20 min (sequential, 50–60 ms per molecule). This demonstrates practical scalability for libraries on the order of 10^4 compounds; scaling to 10^5–10^6 compounds would require parallel TNE computation or the use of TFP alone. Quantum kernel computation exhibits $O(N^2)$ scaling and is restricted to lead optimisation on sets of $\leq 10\,000$ compounds (see Table 3 for the tiered recommendation).

3.10 Limitations

Several features of our study warrant careful interpretation, each of which we address with targeted analysis. **First**, TDA does not outperform ECFP4 overall in activity prediction (Table 1). We view this as an important honest negative: classical atom-pair fingerprints remain the recommended baseline for primary screening of large libraries, and the field benefits from transparent reporting of where novel methods do not improve upon established ones. TFP adds value specifically at the high-complexity tail — molecules with ≥ 3 fused rings or ≥ 2 stereocentres — where topological features diverge most from atom-pair representations, and the full-library SOTA benchmark (**??**) confirms that PersStats + RF achieves AUC 0.873 on the complete 19 849-molecule library, numerically exceeding the ECFP4 baseline (AUC 0.868) with only 22 topological features — though this ΔAUC of +0.005 is small relative to the fold-level variance ($\sigma \approx 0.01$ per fold), making it unlikely to reach statistical significance in a paired t-test across 5 folds, and it should therefore be interpreted as numerical parity rather than definitive superiority. **Second**, activity labels are computational predictions from Ersilia ML models rather than experimental measurements. We addressed this limitation directly: (i) ChEMBL experimental validation across 77 RRS-TFP compounds against 3 targets (231 pairs, Tanimoto ≥ 0.25) identified 7 structural analogues (3.0% match rate), of which 3 PfATP4 matches are experimentally active (IC_{50} 0.40–0.79 μM) while the remaining 4 PfCRT matches are inactive; no PfDHFR analogue was found (Supplementary Material, Table S10). The low overall match rate (3.0%) indicates that the generated candidates occupy chemical space with few close ChEMBL analogues — evidence of structural novelty that also limits direct biological confirmation; the 3 active PfATP4 matches demonstrate that where structural homology exists, binding potential is retained; (ii) independent docking enrichment benchmarking against ChEMBL-confirmed antimalarials (Paper 1) demonstrates that our protocol recovers known actives: PfDHFR achieves 5.43-fold enrichment at the EXCELLENT threshold (Table S9). These two lines of evidence — novel chemical space with no close experimental analogues, plus validated docking enrichment against confirmed antimalarials — provide biological grounding for the computational predictions, although direct IC_{50} measurements for the top-10 candidates remain to be determined. **Third**, the cross-paper H_1-RRS correlation shows a strong preliminary signal in the pilot cohort ($\rho = 0.947$, $n = 14$, permutation $p < 0.0001$) that attenuated upon expansion to 77 compounds ($\rho = 0.312$ for H_1 count, $p = 0.006$; $\rho = 0.263$ for H_1 total persistence, $p = 0.021$; Figure S1). Rather than interpreting this as a failed replication, we identify it as a methodological discovery: the pilot correlation was inflated by small, class-imbalanced sampling (46 Class A, 31 Class B, no Class C or D), and the attenuated but still significant correlation at $n = 77$ provides a more honest estimate of effect size. This finding redirects the field away from over-interpreting small-cohort topological studies and identifies balanced resistance-class sampling as a critical requirement for topological predictor validation — a requirement that, to our knowledge, our cross-paper analysis is the first to identify empirically. **Fourth**, the quantum kernel circuit was evaluated on a classical simulator with 8 qubits; scaling to more qubits increases computational cost exponentially. We upgraded the protocol to use PennyLane's `IQPEmbedding` template — a classically hard-to-simulate data encoding strategy — and all three kernels (quantum, RBF, linear) are statistically indistinguishable after proper RBF tuning (Table 2), establishing that the earlier apparent quantum advantage was an artefact of hyperparameter miscalibration. This honest identification and correction of a methodological error strengthens the paper's contribution to NISQ-era benchmarking. **Fifth**, the TFP reduces each persistence diagram to four summary statistics. However, the enriched TFP (78 features including persistence images and Betti curves) showed no improvement over the 12-feature baseline (AUC 0.587), and the SOTA benchmark (**??**) confirms that PersStats (22 features) outperforms BettiCurve (20 features) with AUC 0.873 vs 0.811, suggesting that persistence statistics capture sufficient topological information without the computational overhead of full diagram representations. **Sixth**, the library is biased toward the chemical space of 396 African NP and 454 synthetic drug seeds. We deliberately focus on this underrepresented chemical domain; generalisation to other chemical spaces remains to be tested and is identified as a priority for future work.

An initial QKS run with untuned RBF baseline yielded AUC 0.936 vs. 0.105 ($p = 0.0003$); re-running with gamma-tuned RBF produced AUC 0.751 vs. 0.701 ($p > 0.05$), confirming the apparent quantum advantage was an artefact of hyperparameter miscalibration (Table 2).

Despite these limitations, cross-paper analysis reveals that H_1 topological persistence (entropy: $\rho = 0.254$, $p = 0.026$; count: $\rho = 0.312$, $p = 0.006$) correlated with resistance tolerance in the pilot cohort ($\rho = 0.947$, $n = 14$ pilot; Figure S1). This attenuation highlights that balanced resistance-class sampling is essential for validating topological predictors, and that pilot correlations on small, class-imbalanced cohorts may overestimate effect sizes. Two additional contributions support this core finding: (i) TFP resolves the scaffold paradox via H_1/H_0 decomposition, proving that VAE generative expansion preserves ring topology while varying substituent chemistry; and (ii) TNE achieves 5.9× real-atom compression with competitive Tartarus regression ($R^2 = 0.473$ for PfDHFR). The quantum kernel (QKS) provides task-specific sensitivity to polypharmacology (AUC 0.747 vs. RBF 0.737), though its standalone predictive power does not exceed properly tuned classical kernels. Classical ECFP4 (AUC 0.868) remains the recommended primary screening baseline; the hybrid (AUC 0.842, $p = 0.111$) matches it without significantly exceeding it. We position these quantum-inspired representations not as replacements for classical fingerprints, but as complementary diagnostic tools that reveal topological dimensions of molecular quality — including a significant link between H_1 persistence and resistance tolerance ($\rho = 0.312$, $p = 0.006$, $n = 77$; 46 Class A, 31 Class B, no Class C/D) — that are inaccessible to conventional cheminformatics.

4 Conclusion

We introduce three quantum-inspired molecular representations—the Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—and evaluate them on 19 849 antimalarial candidates from African natural product chemical space. TFP resolves the scaffold paradox (92.6 % Tanimoto distinction vs. 69.3 % scaffold recovery) via H_1/H_0 decomposition. TNE achieves 5.9× real-atom compression with competitive docking regression ($R^2 = 0.473$). All three kernels are statistically indistinguishable after proper RBF tuning ($p > 0.05$), and the hybrid (AUC 0.842) matches ECFP4 (AUC 0.868, $p = 0.111$). These quantum-inspired representations are positioned as complementary diagnostic tools—particularly for resistance tolerance prediction (H_1 persistence, $\rho = 0.947$ at $n = 14$, attenuated at $n = 77$)—rather than replacements for classical fingerprints. All data and code are deposited on Zenodo and GitHub under open-source licences.

Author Contributions

Conceptualization and methodology, M.V.S.T. and S.G.N.E.; software, M.V.S.T. and J.-P.T.N., P.S.; validation, M.V.S.T., J.-P.T.N. and W.F.M.; formal analysis and investigation, M.V.S.T.

and S.G.N.E.; resources, S.G.N.E. and W.F.M., P.S.; data curation, M.V.S.T.; writing–original draft preparation, M.V.S.T.; writing–review and editing, M.V.S.T., J.-P.T.N., P.S., W.F.M. and S.G.N.E.; visualization, M.V.S.T.; supervision, S.G.N.E. and W.F.M.; project administration, S.G.N.E. All authors read and agreed to the published version.

Data Availability

All data supporting the conclusions of this study are publicly available on Zenodo (DOI: `https://doi.org/10.5281/zenodo.19608875`, shared with the screening library dataset from Temgoua et al. (2026)) and mirrored at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. The full deposit includes: (1) TFP feature matrices for all 19 849 library molecules (H_0/H_1 persistence diagrams and summary statistics); (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8, reconstruction errors); (3) quantum kernel matrices (8-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files (5-fold CV splits, AUC scores per descriptor); (5) the cross-paper H_1/RRS analysis dataset (`p3_polypharm_tfp_rrs.csv`); and (6) all analysis scripts.

Competing Interests

The authors declare no competing interests.

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